GWAS Lab 6

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CS-4643 Bioinformatics

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A. Chi-Square / Fisher Exact Test

The toy GWAS dataset contained **89 subjects** and **17 SNP predictors**. For SNP **rs630969**, the heterozygous genotype counts were **17** for phenotype 0 and **21** for phenotype 1.

The observed contingency table for **rs634228** was:

Table 1: Observed contingency table for rs634228

phenotype	AA	AG	GG
0	0	12	29
1	1	10	37

For rs634228, the genotype marginal totals were **(1, 22, 66)**, the phenotype marginal totals were **(41, 48)**, and the total number of subjects was **89**.

The expected contingency table for rs634228 was:

Table 2: Expected contingency table for rs634228

phenotype	AA	AG	GG
0	0.4606742	10.134831	30.404494
1	0.5393258	11.865169	35.595506

Fisher's exact test identified **rs7835221** as the strongest SNP, followed by **rs2460911** and **rs2460915**. There were **3 SNPs significant at FDR** < 0.05 and **2 SNPs significant at Bonferroni** < 0.05.

Fisher exact test code

```
observed_tables_list01 <- lapply(genotypes_df, function(x) {
  table(phenotype01, x)
```

```
})

fisher_results <- tibble(
  SNP = names(observed_tables_list01),
  p = sapply(observed_tables_list01, function(tab) fisher.test(tab)$p.value)
) %>%
  arrange(p) %>%
  mutate(
    p_adj_fdr = p.adjust(p, method = "fdr"),
    p_adj_bonf = p.adjust(p, method = "bonferroni")
  )

print(head(fisher_results, 10))
```

Top Fisher results

SNP	p	p_adj_fdr	p_adj_bonf
rs7835221	1.578661e-13	2.68e-12	2.68e-12
rs2460911	8.032123e-04	6.83e-03	1.37e-02
rs2460915	3.925298e-03	2.22e-02	6.67e-02
rs529983	1.838391e-01	6.73e-01	1.00e+00
rs6999231	2.096815e-01	6.73e-01	1.00e+00
rs17786052	2.375773e-01	6.73e-01	1.00e+00
rs12156420	2.919794e-01	7.09e-01	1.00e+00
rs17121574	4.033435e-01	7.12e-01	1.00e+00
rs10105623	4.193435e-01	7.12e-01	1.00e+00
rs2460914	4.448165e-01	7.12e-01	1.00e+00

B. Logistic Regression

SNP 8 corresponded to **rs7835221**. Logistic regression showed that **rs7835221** was the strongest signal. The estimated coefficients were **2.8904** for the intercept, **-3.4369** for AG, and **-5.8861** for GG. The corresponding p-values were **0.0000693**, **0.0000273**, and **0.00000278**.

The genotype with the highest disease susceptibility was **AA**, because AA had many more cases than controls, while AG and GG had substantially reduced odds relative to AA.

The logistic regression ranking was similar to Fisher's exact test for the strongest SNPs, especially **rs7835221**. Lower-ranked SNPs were less stable because some genotype categories were sparse.

Logistic regression code

```
pheno_factor <- factor(phenotype01, levels = c(0, 1))
```

```
i <- 8
lr <- glm(pheno_factor ~ genotypes_df[[i]], family = binomial)

summary(lr)
td_lr <- tidy(lr)

td_lr
colnames(genotypes_df)[i]
table(genotypes_df[[i]], phenotype01)
```

Logistic regression output for SNP 8

Table 4: Logistic regression output for rs7835221

term	estimate	std.error	statistic	p.value
(Intercept)	2.8904	0.7265	3.979	0.0000693
genotypes_df[[i]]AG	-3.4369	0.8193	-4.195	0.0000273
genotypes_df[[i]]GG	-5.8861	1.2560	-4.686	0.00000278

SNP 8: rs7835221

Genotype counts for rs7835221

Table 5: Genotype counts for rs7835221

genotype	phenotype 0	phenotype 1
AA	2	36
AG	19	11
GG	20	1

Logistic regression function and ranked output

```
LR_fn <- function(i){

  lr <- glm(pheno_factor ~ genotypes_df[[i]], family = binomial)

  td_lr <- tidy(lr)

  pval_vec <- td_lr$p.value
  coef_vec <- td_lr$estimate
```

```r
  cbind(
    SNP = colnames(genotypes_df)[i],
    b0 = coef_vec[1],
    b1 = coef_vec[2],
    b2 = coef_vec[3],
    p0 = pval_vec[1],
    p1 = pval_vec[2],
    p2 = pval_vec[3]
  )
}

lr_results <- lapply(1:ncol(genotypes_df), LR_fn)
lr_results_df <- do.call(rbind, lr_results)
lr_results_df <- as.data.frame(lr_results_df, stringsAsFactors = FALSE)

lr_results_df$p2 <- as.numeric(lr_results_df$p2)
lr_results_df <- lr_results_df[order(lr_results_df$p2), ]

head(lr_results_df, 10)
```

Top logistic regression results

SNP	b0	b1	b2	p0	p1	p2
rs7835221	2.89037	-3.43692	-5.88610	6.93e-05	2.73e-05	2.78e-06
rs607499	-0.18232	0.40547	0.55005	0.67025	0.43802	0.36674
rs630969	0.26826	-0.05695	-0.36357	0.46654	0.90786	0.52469
rs10105623	8.05e-15	-0.22314	0.34831	1.00000	0.83516	0.73646
rs2460915	0.72392	-1.06039	-17.28999	0.01749	0.01980	0.98850
rs17786052	-16.56607	16.56607	16.81119	0.99046	0.99046	0.99032
rs2460911	0.75377	-1.05915	-18.31984	0.01291	0.02266	0.99095
rs529983	-15.56607	15.09606	15.83800	0.99147	0.99172	0.99132
rs17121574	-15.56607	14.87292	15.77863	0.99147	0.99185	0.99135
rs11203962	-15.56607	15.56607	15.77579	0.99147	0.99147	0.99135

C. Manhattan Plot, QQ Plot, and LASSO

The Manhattan plot and QQ plot both highlighted **rs7835221** as the strongest association signal in the toy dataset. The genomic inflation factor was $\lambda = 1.433506$, indicating moderate inflation above the null expectation, likely driven by a small number of strongly associated SNPs.

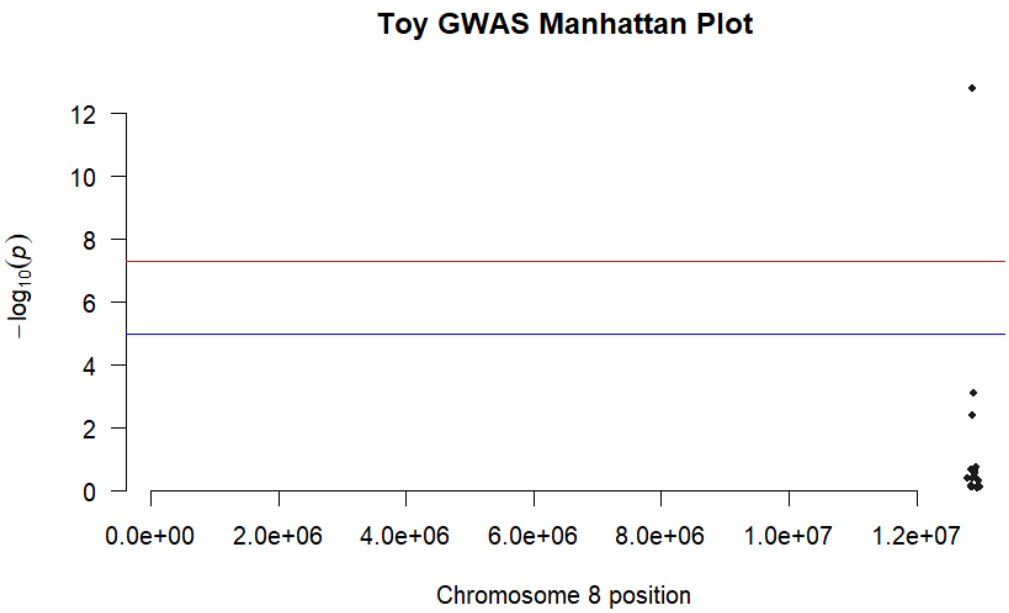

Figure 1: Manhattan plot for the toy GWAS dataset. The strongest signal was observed at **rs7835221**, which had the smallest Fisher's exact test p-value.

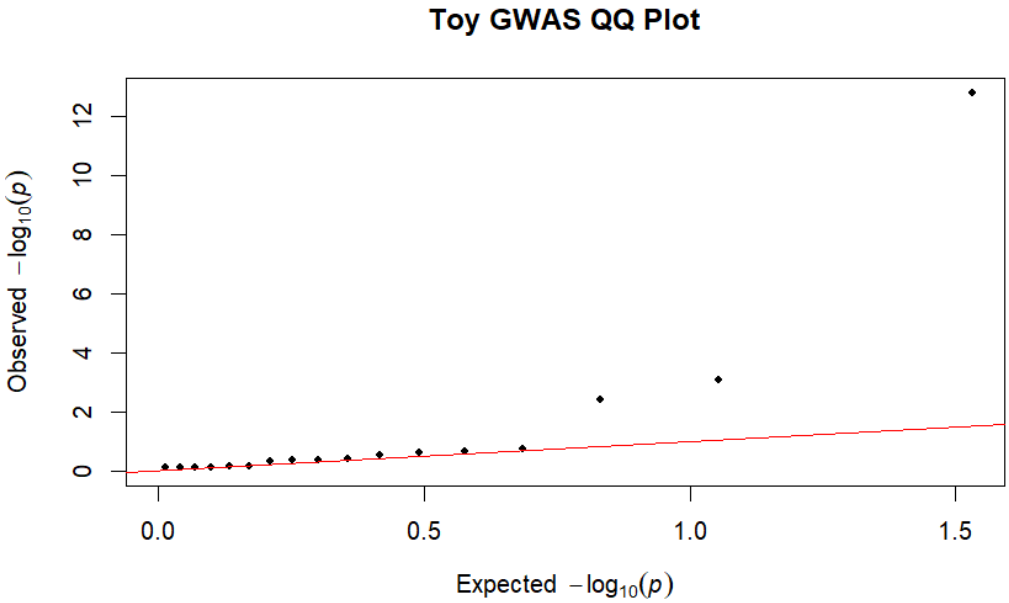

Figure 2: QQ plot for the toy GWAS dataset. Most SNPs followed the null expectation, while a small number of SNPs deviated upward, especially **rs7835221**, indicating strong association signal.

In the LASSO model, **rs7835221** was the only SNP retained at `lambda.1se`, showing that it was

the strongest multivariate predictor in the toy dataset.

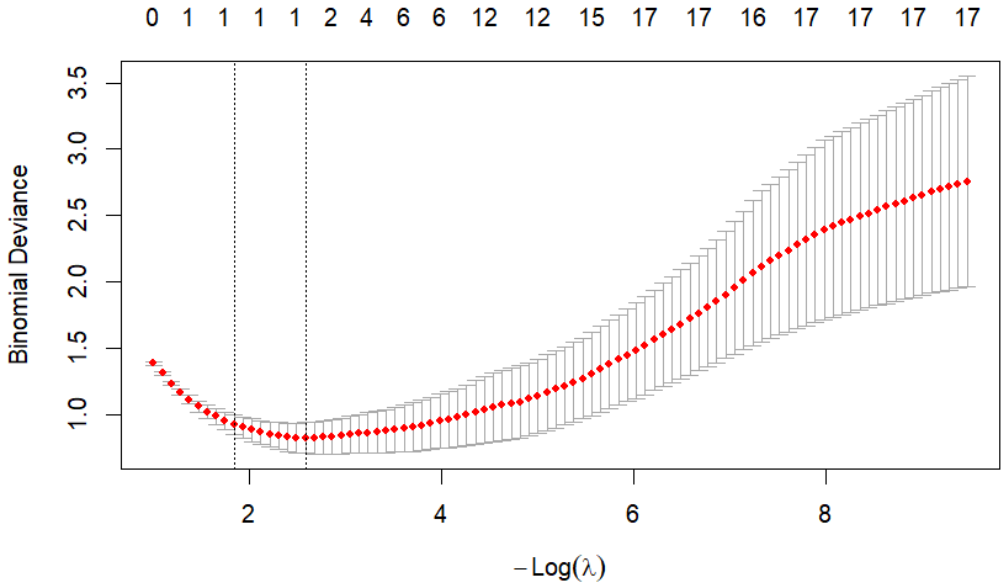

Figure 3: Cross-validation curve for the LASSO logistic regression model. At `lambda.1se`, **rs7835221** was the only SNP retained in the toy dataset.

Manhattan / QQ code

```
toyResults <- data.frame(
  CHR = extra_map$CHR,
  SNP = extra_map$SNP,
  BP = extra_map$BP,
  P = sapply(observed_tables_list01, function(tab) fisher.test(tab)$p.value)
)

toyResults$CHR <- as.numeric(toyResults$CHR)
toyResults$BP <- as.numeric(toyResults$BP)
toyResults$P <- as.numeric(toyResults$P)

toyResults

manhattan(toyResults, main = "Toy GWAS Manhattan Plot")
qq(toyResults$P, main = "Toy GWAS QQ Plot")

chisq_stats <- qchisq(1 - toyResults$P, df = 1)
lambda <- median(chisq_stats, na.rm = TRUE) / qchisq(0.5, df = 1)
lambda
```

toyResults

CHR	SNP	BP	P
8	rs17121574	12799052	4.033435e-01
8	rs754238	12848056	7.830830e-01
8	rs11203962	12848438	6.871138e-01
8	rs6999231	12862253	2.096815e-01
8	rs17178729	12867001	7.271907e-01
8	rs10105623	12868315	4.193435e-01
8	rs2460915	12870407	3.925298e-03
8	rs7835221	12878098	1.578661e-13
8	rs2460911	12895289	8.032123e-04
8	rs12156420	12914557	2.919794e-01
8	rs17786052	12922389	2.375773e-01
8	rs529983	12942555	1.838391e-01
8	rs630969	12945844	7.961562e-01
8	rs2460914	12958068	4.448165e-01
8	rs607499	12961886	7.237858e-01
8	rs634228	12963283	4.609209e-01
8	rs556531	12989321	7.476514e-01

Genomic inflation factor: 1.433506

LASSO code and output

```
genotypes_num <- sapply(genotypes_df, function(x) {
  ux <- sort(unique(na.omit(x)))
  as.numeric(factor(x, levels = ux)) - 1
})

genotypes_num <- as.matrix(genotypes_num)
genotypes_num[is.na(genotypes_num)] <- 0

colnames(genotypes_num) <- colnames(genotypes_df)

cvfit <- cv.glmnet(
  x = genotypes_num,
  y = phenotype01,
  family = "binomial",
  alpha = 1
)

plot(cvfit)
```

```r
lasso_coef <- coef(cvfit, s = "lambda.1se")
lasso_coef

lasso_mat <- as.matrix(lasso_coef)
selected_snps <- rownames(lasso_mat)[lasso_mat[,1] != 0]
selected_snps <- setdiff(selected_snps, "(Intercept)")
selected_snps
```

LASSO coefficients

Table 8: LASSO coefficients at lambda.1se

Variable	Coefficient
(Intercept)	1.113710
rs7835221	-1.166541

Selected SNP at lambda.1se: rs7835221

D. Benchmark Simulated Datasets

Dataset 1: `a_20s_1600her_0.1__maf_0.2_EDM-2_01.txt`

The first benchmark dataset had **1600 samples** and **21 variables**, with balanced classes (**800 controls** and **800 cases**). Univariate logistic regression identified **N7** as the strongest marginal predictor (`p = 0.00277`). Random forest ranked **M0P0** highest, followed by **N14**, **N1**, **N8**, and **N13**. LASSO selected no predictors at `lambda.1se`, but selected **N7** at `lambda.min`. This suggests that **N7** is the clearest additive signal, while other methods captured additional structure.

Dataset 2: `a_20s_1600_Het_h_0.4MAF_0.2_r_50_EDM-2_01.txt`

The second benchmark dataset also had **1600 samples** and **21 variables**, with balanced classes (**800 controls** and **800 cases**). Univariate logistic regression showed weak marginal effects, with **N12** as the top feature (`p = 0.0545`). Random forest ranked **M0P0**, **M1P1**, **M1P0**, and **M0P1** highest. LASSO selected no predictors at either `lambda.1se` or `lambda.min`. This suggests that the second dataset may contain interaction-driven or nonlinear structure rather than strong additive main effects.

Method comparison

Across methods, the toy dataset showed strong agreement, with **rs7835221** identified consistently by Fisher's exact test, logistic regression, Manhattan/QQ analysis, and LASSO. In contrast, the

benchmark datasets showed weaker agreement, especially Dataset 2, where random forest identified interaction-style variables that were not strongly detected by logistic regression or LASSO.

Benchmark dataset code

```
sim1 <- read.table("a_20s_1600her_0.1__maf_0.2_EDM-2_01.txt", header = TRUE, check.names
    = FALSE)
sim2 <- read.table("a_20s_1600_Het_h_0.4MAF_0.2_r_50_EDM-2_01.txt", header = TRUE, check.
    names = FALSE)

find_class_col <- function(df) {
  possible <- c("Class", "class", "CLASS", "Y", "y", "Outcome", "outcome")
  hit <- intersect(possible, names(df))
  if (length(hit) > 0) return(hit[1])
  return(names(df)[ncol(df)])
}

class_col1 <- find_class_col(sim1)
class_col2 <- find_class_col(sim2)

run_uni_logit <- function(df, class_col) {
  preds <- setdiff(names(df), class_col)

  res <- lapply(preds, function(v) {
    fit <- glm(as.formula(paste(class_col, "~", paste0("`", v, "`"))),
               data = df, family = binomial)
    td <- tidy(fit)
    out <- td[2, c("term", "estimate", "p.value")]
    out$term <- v
    out
  })

  bind_rows(res) %>% arrange(p.value)
}

run_rf <- function(df, class_col) {
  df[[class_col]] <- as.factor(df[[class_col]])
  rf <- randomForest(as.formula(paste(class_col, "~ .")), data = df, importance = TRUE)
  imp <- importance(rf)
  imp_df <- data.frame(
    Feature = rownames(imp),
    MeanDecreaseGini = imp[, "MeanDecreaseGini"],
    row.names = NULL
  )
  imp_df[order(-imp_df$MeanDecreaseGini), ]
}

run_lasso <- function(df, class_col) {
```

```r
  y <- df[[class_col]]
  X <- as.matrix(df[, setdiff(names(df), class_col)])

  cv <- cv.glmnet(X, y, family = "binomial", alpha = 1)
  cm <- coef(cv, s = "lambda.1se")
  cm_mat <- as.matrix(cm)
  sel <- rownames(cm_mat)[cm_mat[,1] != 0]
  setdiff(sel, "(Intercept)")
}

lasso_min <- function(df, class_col) {
  y <- df[[class_col]]
  X <- as.matrix(df[, setdiff(names(df), class_col)])

  cv <- cv.glmnet(X, y, family = "binomial", alpha = 1)
  cm <- coef(cv, s = "lambda.min")
  cm_mat <- as.matrix(cm)
  sel <- rownames(cm_mat)[cm_mat[,1] != 0]
  setdiff(sel, "(Intercept)")
}

uni1 <- run_uni_logit(sim1, class_col1)
uni2 <- run_uni_logit(sim2, class_col2)

rf1 <- run_rf(sim1, class_col1)
rf2 <- run_rf(sim2, class_col2)

lasso1 <- run_lasso(sim1, class_col1)
lasso2 <- run_lasso(sim2, class_col2)

lasso1_min <- lasso_min(sim1, class_col1)
lasso2_min <- lasso_min(sim2, class_col2)
```

Dataset 1 dimensions and class balance

Dimensions: **1600 x 21**
Class balance: **800 controls, 800 cases**

Dataset 2 dimensions and class balance

Dimensions: **1600 x 21**
Class balance: **800 controls, 800 cases**

Top univariate logistic regression results for Dataset 1

term	estimate	p.value
N7	0.366	0.00277
N17	-0.0910	0.207
N6	0.0946	0.370
N12	-0.114	0.372
N2	-0.144	0.375
N0	0.0653	0.419
N3	-0.0644	0.422
N10	-0.0557	0.445
N11	0.0492	0.483
N5	-0.0423	0.561

Top univariate logistic regression results for Dataset 2

term	estimate	p.value
N12	0.165	0.0545
N0	-0.165	0.152
N6	0.0957	0.231
M1P1	0.103	0.248
N15	0.147	0.280
N4	0.0828	0.282
N3	0.241	0.300
N9	-0.0669	0.391
N1	-0.0594	0.409
N14	0.0833	0.499

Top random forest results for Dataset 1

Feature	MeanDecreaseGini
M0P0	47.00207
N14	44.92108
N1	44.61101
N8	44.47669
N13	44.19554
N5	44.08519
N9	44.05866
N10	43.75840
N11	43.01474

M0P1	42.74257

Top random forest results for Dataset 2

Feature	MeanDecreaseGini
M0P0	52.62911
M1P1	50.63148
M1P0	50.45973
M0P1	47.62529
N8	45.10102
N2	44.30587
N1	43.78821
N5	42.28909
N4	40.97747
N9	40.66679

LASSO outputs

- **Dataset 1, lambda.1se:** no predictors selected
- **Dataset 1, lambda.min:** N7
- **Dataset 2, lambda.1se:** no predictors selected
- **Dataset 2, lambda.min:** no predictors selected

Conclusion

Overall, the toy dataset showed a strong and consistent signal at **rs7835221**, which was identified by Fisher's exact test, logistic regression, Manhattan/QQ analysis, and LASSO. The benchmark datasets showed weaker agreement across methods. Dataset 1 contained a clearer additive signal at **N7**, while Dataset 2 appeared to be driven more by nonlinear or interaction-based structure, especially based on the random forest rankings of the M/P variables.

Overall, these results show that different GWAS analysis methods can agree strongly when there is a clear additive signal, but may diverge when the underlying structure is more nonlinear or interaction-driven.